

Bank Loan Analysis — Project Report

End-to-end data analytics project: clean → analyze → visualize. Prepared
by Abhishek Anthony · Version 1.0 · July 2026



Executive Summary

Purpose

Analyze loan applications and build interactive Power BI dashboards to support data-driven lending decisions.

Workflow

Data cleaning (Python) → SQL analysis (MySQL) → DAX measures → Power BI dashboards.

Dashboards

Four pages: Executive Overview, Loan Performance, Customer Analysis, Geographic Analysis.



Problem & Business Questions

As loan volumes grow, manual reporting fails to reveal risk patterns. An interactive BI solution is required to monitor KPIs and support fast decisions.

- How many applications? Total funded amount? Payments received?
- Top loan grades, purposes, customer segments, and states?
- Distribution: Fully Paid / Current / Charged Off?



Objectives & Scope

1

Data Preparation

Collect, clean missing/duplicate values, validate types using Python (Pandas, NumPy).

2

SQL Analysis

Store cleaned data in MySQL; compute KPIs with aggregate, CTEs, and window functions.

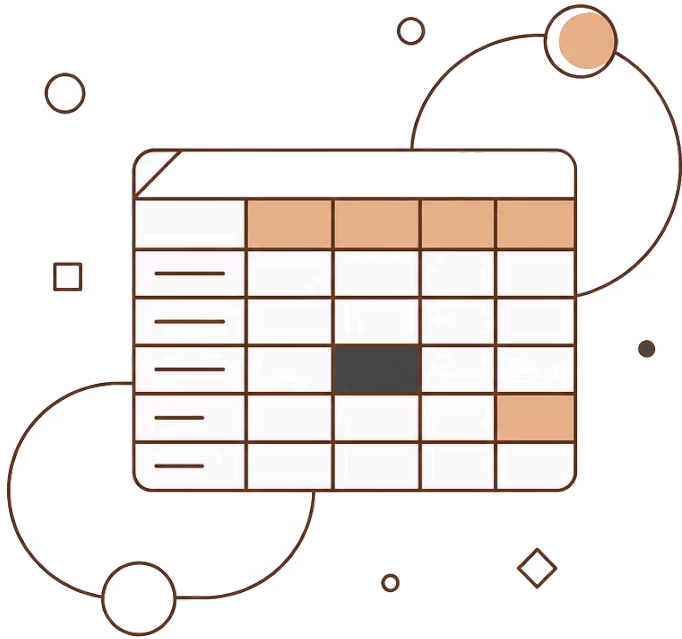
3

Dashboard Development

Design Power BI pages, build DAX measures and interactive visuals for business users.



Dataset & Tools



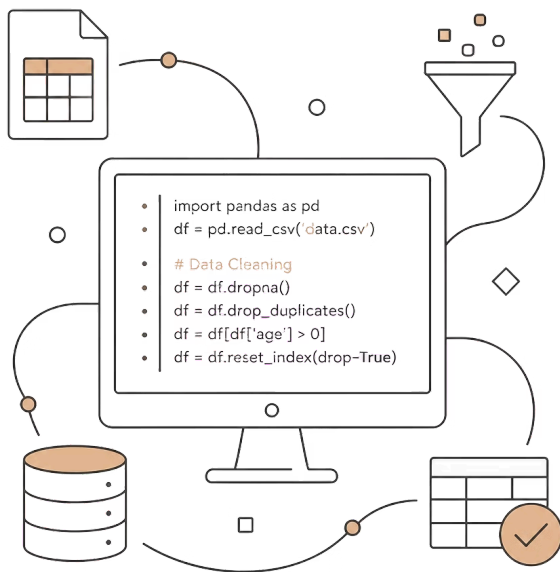
Dataset

Realistic loan dataset with financial, customer, and geographic attributes—suitable for end-to-end analysis.

Tools

- Excel, Python (Pandas, NumPy), MySQL, SQL
- Power BI, DAX, Microsoft Word, Gama AI

Data Cleaning & Preprocessing



01

Load & Inspect

`head()`, `info()`, `describe()`, `shape` used to understand structure.

02

Clean

Missing values handled, duplicates removed, types validated (dates, numeric fields).

03

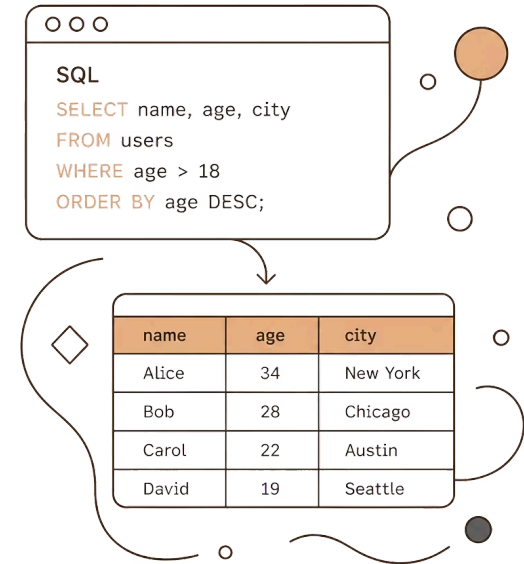
Outliers & Export

Outlier checks performed; final cleaned dataset exported for MySQL.

SQL Analysis — Approach & KPIs

SQL queries answered business questions and produced KPIs used in dashboards.

- Total Loan Applications, Total Funded Amount, Total Payment Received
- Average Interest Rate, Avg Annual Income, Avg DTI
- Loan status, grade performance, state-wise funding



Power BI — Dashboards & Features



Executive Overview

KPI cards, trend lines, and high-level maps for monitoring portfolio health.



Loan Performance

Funding vs repayments, loan-status distribution, grade comparisons.



Customer Analysis

Home ownership, income vs loan amount, occupation-wise lending.



Geographic Analysis

State-wise funded amounts, payments received, and regional trends.

Interactive features: slicers, drill-downs, cross-filtering, navigation buttons, and responsive KPI cards.

Key Business Insights & Recommendations

- **Insights**

- Active lending with significant funded amount and payments received.
- Loan status mix: Fully Paid, Current, Charged-Off — monitor charged-off trends.
- Borrowing varies by home ownership, income, purpose, and region.
- Interest rates and loan grades highlight risk and pricing opportunities.

- **Recommendations**

1. Strengthen credit risk assessment (DTI, grade monitoring).
2. Target high-performing customer segments with tailored products.
3. Develop region-specific lending strategies; monitor charged-offs.
4. Implement real-time dashboards and consider predictive models for defaults.



Conclusion & Future Scope

Project achieved an end-to-end pipeline: Python cleaning → MySQL analysis → Power BI dashboards with DAX. Delivered actionable insights and recommendations to improve lending decisions.

Future Enhancements

- Predictive models for default probability (Logistic Regression, XGBoost).
- Real-time data refresh and cloud deployment (Azure / AWS / GCP).
- AI-driven executive summaries, automated risk alerts, and system integration with core banking.



📄 Prepared using theme colors #3E2513 and #F3E3D8. Contact: Abhishek Anthony for full report and Power BI files.